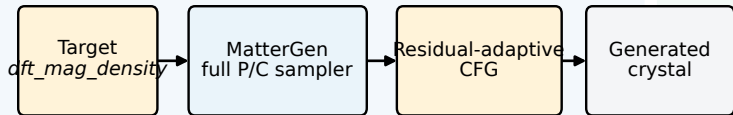
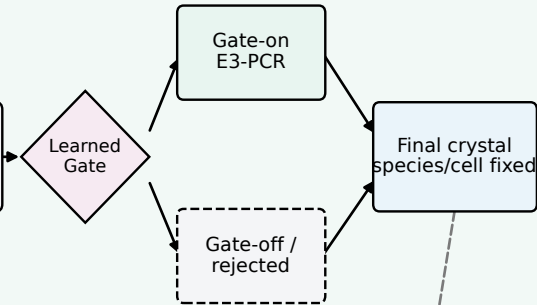


Innovation 1 · sampling-stage control

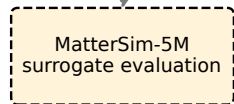


Innovation 2 · safe post-generation refinement

5 bounded position steps



Exact fallback to input



Evaluation boundary: surrogate force, relaxation, E-hull and stability; no DFT verification